INL1, Problem 7

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MAA155

$$S\text{-pol}(x+y, xy) = \frac{lc(xy)\cdot t}{lm(x\cdot y)}\cdot(x+y) - \frac{lc(x+y)\cdot t}{lm(x\cdot y)}\, xy$$

with $t = lcm\big(lm(x+y), lm(xy)\big)$

$$lc(x+y) = 1,\quad lc(x\cdot y) = 1$$

$$lm(x+y) = x,\quad lm(xy) = xy$$

so $t = lcm(x, xy) = xy$

$$\text{So } Spol(x+y, xy) = \frac{1\cdot xy}{x}(x+y) - \frac{1\cdot xy}{xy}xy$$

$$= y(x+y) - xy = y^2$$

(ii) we already have $S(x+y, xy) = y^2 \xrightarrow[G]{} 0$

we still need to check that $Spol(x+y, y^2) \xrightarrow[G]{} 0$ and $Spol(xy, y^2) \xrightarrow[G]{} 0$

$$Spol(x+y, y^2) = \frac{xy^2}{x}(x+y) - \frac{xy^2}{y^2}y^2 = y^3 \xrightarrow[G]{} 0 \text{ because } y^2 \xrightarrow[G]{} 0$$

$Spol(xy, y^2) = y\cdot xy - x\cdot y^2 = 0 \xrightarrow[G]{} 0$ So G is a Gröbner bases

(iii) we can see that $xy = (x+y-y)y \xrightarrow[G]{} (0-y)\cdot y = -y^2 \xrightarrow[G]{} 0$

So the new base reduced is $G' = \{x+y, y^2\}$ (which can not be more minimal since $x+y \xrightarrow[G'\setminus\{x+y\}]{} 0$ and $y^2 \xrightarrow[G'\setminus\{y^2\}]{} 0$)

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#EL OTMANI Youssef
#1
R = PolynomialRing(QQ, ['x', 'y'], order='lex')
#x, y = R.gens()
x=var('x')
y=var('y')


f = R(x^2*y + y^3 - 8*y)
g = R(x^2 - y^2)

#2
p1 = implicit_plot(f, (x, -5, 5), (y, -5, 5), color='blue')
p2 = implicit_plot(g, (x, -5, 5), (y, -5, 5), color='red')
show(p1 + p2)

#3 - solution visually  = (0,0) (2,2) (-2,-2) (2,-2) (-2,2)

#4
def spol(f, g):
    t = lcm(f.lm(), g.lm())
    return ((g.lc() * t * f) // f.lm()) - ((f.lc() * t * g) // g.lm())

#5
h = spol(f, g)
print("S-polynomial:", h, "but G is generated by", f, "and", g)

h_reduced = h.reduce([f, g])
print("h reduced:", h_reduced)

#6
h1 = spol(f, h)
h2 = spol(g, h)

h1_reduced = h1.reduce([f, g, h])
h2_reduced = h2.reduce([f, g, h])

print("h1 reduced:", h1_reduced)
print("h2 reduced:", h2_reduced)
# we can see that h1 and h2 can be reduced using h, f and g

G = ideal([f, g, h]).groebner_basis()
print("Gröbner basis:", G)

#7
for poly in G:
    if poly.variables() == (y,):
        poly_y = poly
        break

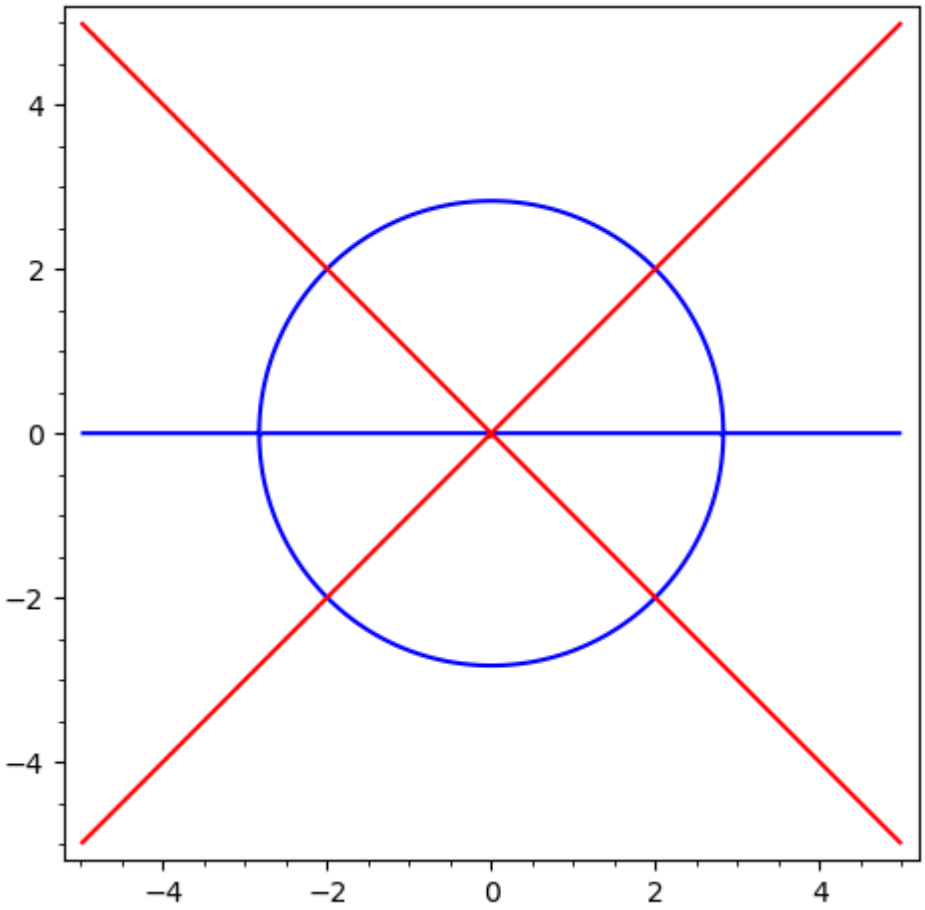
factored_poly_y = poly_y.factor()
print("Factorized polynomial in y:", factored_poly_y)


eq = y * (y - 2) * (y + 2) == 0
y_solutions = solve(eq, y)
print("Solutions for y:", y_solutions)

x_solutions = []
for sol in y_solutions:
    y_val = sol.rhs()  # .rhs() pour obtenir la valeur de y
    x_eq = g.subs(y=y_val)  # Substitute y into g(x, y) = 0
    x_solutions.append(solve(x_eq, x))

for i, sol in enumerate(y_solutions):
    print(f"For y = {sol.rhs()}, x solutions are: {x_solutions[i]}")
```

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Out :
S-polynomial: 2*y^3 - 8*y but G is generated by x^2*y + y^3 - 8*y and x^2 - y^2
h reduced: 2*y^3 - 8*y
h1 reduced: 0
h2 reduced: 0
Gröbner basis: [x^2 - y^2, y^3 - 4*y]
Factorized polynomial in y: y * (y - 2) * (y + 2)
Solutions for y: [
y == -2,
y == 0,
y == 2
]
For y = -2, x solutions are: [
x == -2,
x == 2
]
For y = 0, x solutions are: [
x == 0
]
For y = 2, x solutions are: [
x == -2,
x == 2
]
```


4
2
0
−2
−4
−4
−2
0
2
4